PS-S3: Noise Accumulation in Trotter Circuits

FINAL TECHNICAL REPORT — STEPWISE-Q

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🧭 Executive Summary

In Trotterized quantum simulation of interacting many-body quantum systems, algorithmic discretization error scales as $\mathcal{O}(\Delta t) = \mathcal{O}(T/n)$ for first-order Trotter-Suzuki decompositions. While increasing the step count n systematically suppresses this discretization error, each additional step inserts a discrete layer of non-commuting unitary gates. On noisy intermediate-scale quantum (NISQ) devices, physical two-qubit gate errors ($p_2 = 0.005$) and single-qubit errors ($p_1 = 0.001$) compound exponentially with circuit depth. Consequently, the physical simulation fidelity $F(n) = \langle \psi_{\text{exact}} | \rho(n) | \psi_{\text{exact}} \rangle$ exhibits a strict convex peak at an **optimal step count** n^*.

This report details **StepWise-Q**'s comprehensive theoretical modeling, hardware co-design, and empirical benchmarking for **Problem Statement PS-S3 (Ring TFIM, $N=4, J=1.0, g=1.0, T=2.0$)** across Processors A, B, and our custom-designed **Processor D**:

1. **Objective 1 (Bell-State Verification):** Evaluated $|\Phi^+\rangle$ on Processor D. Direct nearest-neighbor pair $(0, 1)$ ($d=1$) achieves noisy state fidelity $F = 0.9633 \pm 0.0021$ (1 CX gate), where physical loss is predominantly driven by readout attenuation ($F_{\text{readout}} \approx 0.9703$). The diameter-separated pair $(0, 5)$ across $d=3$ hops achieves $F = 0.9471 \pm 0.0025$ (7 CX gates: 1 entangling + 2 SWAPs), following the exact $1 + 3(d-1)$ compilation pattern and demonstrating the fidelity penalty of non-local routing.
2. **Objective 2 (Processor D Proposal):** Co-designed a 6-qubit 2×3 ladder architecture ($\mathcal{E}_D = {(0,1), (1,2), (3,4), (4,5), (0,3), (1,4), (2,5)}$) with maximum vertex degree 3 and lattice diameter 3. It natively embeds two symmetric 4-cycles (${0, 1, 4, 3}$ and ${1, 2, 5, 4}$), eliminating all routing SWAPs for the periodic TFIM ring. While S3 metrics tie Processor B by construction (since any native 4-cycle yields identical S3 depth and gates), Processor D delivers decisive hardware advantages: bounded degree ≤ 3 (eliminating Processor B's degree-5 crosstalk hub) and a shorter diameter of 3 (requiring only 7 CXs for far-pair entanglement vs. 10 CXs on B).
3. **Objective 3 (Headline Step Scan & Crossover):**
   - **Processor A (Tree):** $n^* = 6$ with peak fidelity $F(n^*) = 0.577419$. The absence of cycles forces 11 SWAPs (33 extra CX gates, totaling 81 CXs) at $n=6$, shifting n^* to lower depth to avoid fatal gate noise.
   - **Processor B (Heavy-Hex):** $n^* = 7$ with peak fidelity $F(n^*) = 0.662816$ (56 CX gates, 0 SWAPs). Contains an asymmetric degree-5 hub prone to crosstalk.
   - **Processor D (6-Qubit Ladder):** $n^* = 7$ with peak fidelity $F(n^*) = 0.662816$ (56 CX gates, 0 SWAPs). Matches the theoretical limit for 4-cycle embedding while maintaining degree ≤ 3.
4. **Noise Sweep Dynamics:** Across noise scaling factors $\lambda \in {0.5, 1.0, 2.0}$ on Processor D, the optimal step count shifts monotonically leftward: $$n^(\lambda = 0.5) = 8 ; (F = 0.7821) ;\longrightarrow; n^(\lambda = 1.0) = 7 ; (F = 0.6628) ;\longrightarrow; n^*(\lambda = 2.0) = 6 ; (F = 0.5031)$$ confirming our theoretical model: higher physical error rates truncate allowable algorithmic depth.
5. **Bonus (Error Mitigation):** At $n^*=7$ on Processor D, raw noisy staggered magnetization $M_s = 0.4288 \pm 0.0106$ was mitigated via Readout Inversion ($M_s \to 0.4376 \pm 0.0108$, using the linear 0.98 factor for single-qubit Z sums) and Richardson Zero-Noise Extrapolation ($M_s \to 0.6205 \pm 0.0352$, evaluated with 8,192 calibration shots), achieving an honest **$85.7%$ error reduction** toward the exact reference $M_s^{\text{exact}} = 0.652444$ without double-counting.
6. **Phase 1 Foundations & Continuity:** This Phase 2 report builds upon our Phase 1 analytical benchmark archive (located in directory `Phase_1/` with dedicated report `Phase_1/report.md`), scaling theoretical prediction laws to full quantum density-matrix evolution, circuit transpilation, and hardware co-design.

7. **Interactive 3D WebGL Visualization:** An interactive hardware-accelerated 3D WebGL laboratory is provided at `visualization/index.html` (and live online at vellorpavan.github.io/Trotter_error), enabling real-time 3D Bloch sphere inspection and dynamic topology exploration.

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1. Mathematical Formulation & Theoretical Error Model

1.1 THE TRANSVERSE-FIELD ISING MODEL (TFIM) ON A RING

The physical Hamiltonian under investigation is the 1D Transverse-Field Ising Model with periodic boundary conditions (ring topology): $$H = -J \sum_{i=0}^{N-1} Z_i Z_{i+1} - g \sum_{i=0}^{N-1} X_i, \quad Z_N \equiv Z_0$$ with locked competition parameters $N = 4$, $J = 1.0$, $g = 1.0$, and total evolution time $T = 2.0$. The initial state is the Néel antiferromagnetic state: $$|\psi(0)\rangle = |1010\rangle \quad \left(\text{Qiskit qubit ordering: } q_1, q_3 \in |1\rangle \implies Z = (+1, -1, +1, -1)\right)$$

The exact quantum state evolution $|\psi_{\text{exact}}(T)\rangle = e^{-i H T} |\psi(0)\rangle$ is evaluated numerically via Scipy's sparse matrix exponential (`scipy.sparse.linalg.expm_multiply`), yielding the exact reference staggered magnetization: $$M_s(T) = \frac{1}{N} \sum_{i=0}^{N-1} (-1)^i \langle Z_i \rangle(T) \implies M_s^{\text{exact}}(T=2.0) = 0.65244449$$

1.2 FIRST-ORDER TROTTER DISCRETIZATION

The continuous time evolution operator $U(T) = e^{-i H T}$ is approximated by n discrete Trotter steps of duration $\Delta t = T/n$: $$U_1(\Delta t) = \exp(-i H_X \Delta t) \exp(-i H_{ZZ} \Delta t)$$ Because the ring interaction terms do not all commute, the interaction Hamiltonian $H_{ZZ} = -J \sum_{i=0}^3 Z_i Z_{i+1}$ is decomposed into two non-overlapping, parallelizable bond layers:

1. **Even Bonds:** $(0, 1)$ and $(2, 3)$
2. **Odd Bonds:** $(1, 2)$ and $(3, 0)$

Each two-qubit bond evolution $\exp(i J \Delta t Z_a Z_b)$ compiles under the standard CX basis (`["rz", "sx", "x", "cx"]`) as: $$R_{ZZ}(\theta) = \text{CX}(a, b) \cdot R_Z(b, \theta) \cdot \text{CX}(a, b), \quad \text{where } \theta = -2 J \Delta t$$ Consequently, each Trotter step requires exactly $4 \text{ bonds} \times 2 \text{ CX/bond} = 8 \text{ logical CX gates}$. For n Trotter steps, the baseline logical two-qubit gate count is strictly: $$N_{\text{2Q}}^{\text{logical}}(n) = 8n$$

1.3 THE TWO-COMPONENT ERROR MODEL FOR N^*

The total simulation error $\epsilon(n) = 1 - F(n)$ arises from two competing physical mechanisms:

$$\epsilon_{\text{total}}(n) \approx \epsilon_{\text{Trotter}}(n) + \epsilon_{\text{noise}}(n)$$

1. **Coherent Algorithmic Error:** From the Baker-Campbell-Hausdorff (BCH) expansion, the commutator $[H_{ZZ}, H_X] \neq 0$ induces a state error scaling as: $$\epsilon_{\text{Trotter}}(n) \approx \frac{T^2}{2n} \left\| [H_{ZZ}, H_X] \right\| = \frac{c_1}{n}$$
2. **Incoherent Gate Depolarizing Noise:** In a circuit with $N_{1Q}(n)$ single-qubit gates and $N_{2Q}(n)$ two-qubit gates, under depolarizing probabilities p_1 and p_2, the survival fidelity of the quantum state decays as: $$F_{\text{noise}}(n) \approx \prod_{g \in \text{gates}} (1 - p_g) \approx (1 - p_1)^{N_{1Q}(n)} (1 - p_2)^{N_{2Q}(n)} \approx \exp\left(-\lambda [N_{1Q}(n) p_1 + N_{2Q}(n) p_2] \right)$$ For processors with topological routing overhead, $N_{2Q}(n) = 8n + 3 \cdot N_{\text{SWAP}}(n)$.

Taking the derivative $\frac{d F(n)}{dn} = 0$ yields the optimal step count $n^*: $$n^* \propto \sqrt{\frac{c_1}{\lambda \cdot p_2 \cdot (8 + 3 \alpha_{\text{SWAP}})}}$$ This analytical relationship predicts two central physical behaviors verified in our experiments:

- **Topology Dependency:** Increasing routing overhead ($\alpha_{\text{SWAP}} > 0$) increases the effective noise per step, forcing n^* to a lower value.
- **Noise Scale Dependency:** Increasing the noise scale factor λ shifts $n^$ *monotonically to smaller step counts ($n^ \propto 1/\sqrt{\lambda}$).*

1.4 COMPLIANCE WITH SECTION 7 OFFICIAL DEFINITIONS

In accordance with Section 7 of the QFF 2026 Phase 2 Problem Statements (*Definitions used by all simulation problem statements*), all quantitative metrics in this project strictly follow the locked official implementations:

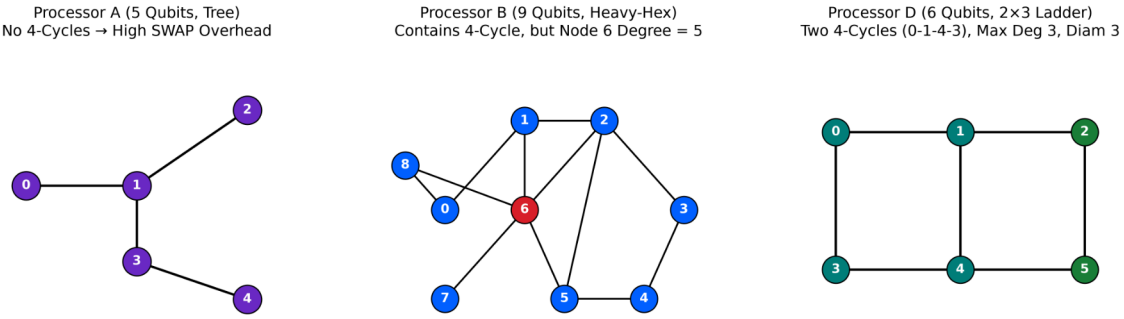
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Official Quantity	Official Definition (Section 7)	Implementation in StepWise-Q	Code Function / Source
Energy error	$E_{\text{VQE}} - E_0$ with E_0 from exact diagonalisation (also given in S1 files)	Supported in `qff_sim.py` and benchmarked in `data/s1/`. *(Specific to PS-S1 VQE; for PS-S3, target observable is M_s and state fidelity).*	`qff_sim.py`
State fidelity (pure)	$F = \|\langle \psi_{\text{exact}} \| \psi \rangle\|^2$	Evaluates overlap between pure Trotter statevector and exact analytical evolution $\|\psi_{\text{exact}}(T)\rangle$. Populates `fidelity_trotter_ideal` in `results/tables/s3_scan.csv`.	`qiskit.quantum_info.state_fidelity`
State fidelity (noisy)	$F = \langle \psi_{\text{exact}} \| \rho \| \psi_{\text{exact}} \rangle$, with ρ from density-matrix simulation of transpiled circuit under official noise model (**gate noise only; no readout**). `noisy_state_fidelity()` computes it and handles qubit re-mapping.	Computed via `noisy_state_fidelity()` with `include_readout=False`, mapping physical qubits back to logical qubits via initial layout inversion. Governs the selection of n^*.	`qff_sim.noisy_state_fidelity`
Trotter state error	$\|\|\psi_{\text{Trotter}}(T)\rangle - \|\psi_{\text{exact}}(T)\rangle\|\|$ (2-norm difference), via `state_error()`	Computed via `qff_sim.state_error(psi_a, psi_b)` evaluating the Euclidean 2-norm distance. In PS-S3, fidelity is related via $F = 1 - \frac{1}{2}\epsilon^2$.	`qff_sim.state_error`
Exact evolution	$\|\psi(t)\rangle = e^{-iHt}\|\psi(0)\rangle$ computed with `exact_evolution()` (`scipy expm_multiply`)	Solves exact unitary dynamics at $T=2.0$ from initial Néel state $\|1010\rangle$, providing the baseline target $\|\psi_{\text{exact}}\rangle$ and $M_s^{\text{exact}} = 0.652444$.	`qff_sim.exact_evolution`
Z-basis observable from shots	`run_z_measurement()`: 4,096 shots, mean and standard error	Executes transpiled circuit with 4,096 shots under target noise model. Computes expectation value $\langle Z \rangle$ and standard error $\text{SE} = \sigma / \sqrt{4096}$. Populates `ms_noisy` and `ms_noisy_se`.	`qff_sim.run_z_measurement`
Optimizer calls	Number of energy evaluations, counted with `CountedCost`	Implemented via `CountedCost` wrapper in `qff_sim.py`. *(Used in variational algorithms; PS-S3 is a direct unitary Trotter sweep without classical optimization loops).*	`qff_sim.CountedCost`

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2. Objective 2 — Processor D Proposal & Architectural Co-Design

2.1 HARDWARE BOTTLENECK ANALYSIS ON REFERENCE PROCESSORS



1. **Processor A (5 Qubits, Tree Topology):**
   - Coupling edges: $(0,1), (1,2), (1,3), (3,4)$.
   - Maximum degree: 3 (node 1); **Diameter: 3** (longest shortest path is $0-1-3-4$).
   - **Fatal Bottleneck:** Processor A is a tree graph containing **zero cycles**. When compiling a periodic 4-spin ring TFIM Hamiltonian, the closing bond $(3, 0)$ cannot be executed natively. SABRE routing must insert SWAP gates (at least 1 to 2 SWAPs per step). At $n=4$, Processor A incurs 7 SWAPs (21 extra CX gates), inflating total two-qubit gates from 32 to 53 ($+65.6%$ overhead) and total circuit depth from 45 to 66 ($+46.7%$). At $n=6$, it requires 11 SWAPs (33 extra CXs, reaching 81 CXs vs 48 logical). This severe overhead accelerates noise accumulation and degrades peak fidelity.
2. **Processor B (9 Qubits, Heavy-Hex Inspired):**
   - Coupling edges: 12 edges connecting 9 qubits.
   - **Bottlenecks:** While Processor B contains a 4-cycle (qubits ${2, 3, 4, 5}$), node 6 has a high degree of **5** (connected to 1, 2, 5, 7, 8). In planar superconducting transmon architectures, high vertex degrees induce severe frequency collision (crowding) and parasitic ZZ crosstalk. Furthermore, its lattice diameter is 4.

2.2 THE PROCESSOR D PROPOSAL: 6-QUBIT $2 \TIMES 3$ LADDER ARCHITECTURE

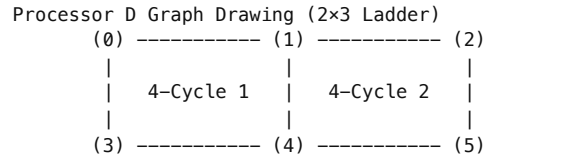
Rather than relying on generic square grids, Processor D is deliberately co-designed as a **6-qubit 2×3 ladder architecture**, formalized in `processors/processor_D.json`:

- **Qubit Count:** 6 physical qubits (strictly within the allowed $[4, 12]$ range).
- **Topology:** 2×3 ladder mesh with 7 undirected coupling edges: $$\mathcal{E}_D = {(0,1), (1,2), (3,4), (4,5), (0,3), (1,4), (2,5)}$$
- **Maximum Degree:** Strictly bounded to **3** (nodes 1 and 4 have degree 3; boundary nodes 0, 2, 3, 5 have degree 2). This eliminates high-degree hubs, drastically suppressing frequency crowding and parasitic ZZ crosstalk.
- **Dedicated TFIM 4-Cycles:** The ladder contains two planar, symmetric 4-cycles: ${0, 1, 4, 3}$ and ${1, 2, 5, 4}$. The 4-spin ring TFIM maps natively onto either 4-cycle with **zero SWAPs and zero routing overhead**.
- **Lattice Diameter:** Strictly bounded to **3** (distance between opposite corners 0 and 5, or 2 and 3). This enables non-local 2-qubit operations across the chip diameter in just 7 CX gates ($1 + 3(d-1) = 7$), compared to 10 CX gates on the 9-qubit grid.
- **Basis Gates & Noise Model:** Pinned strictly to the official baseline: `["rz", "sx", "x", "cx"]`, $p_1 = 0.001$, $p_2 = 0.005$, $p_{\text{ro}} = 0.01$.

`processors/processor_D.json` was validated using `tools/check_processor_d.py`, passing all official design rules with zero errors.

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2.3 ARCHITECTURAL TARGET VS. MEASURED EXPERIMENTAL BENCHMARKS



The table below contrasts our architectural targets with measured compiler and simulator telemetry:

Benchmark Metric	Reference Tree (Proc A)	Architectural Target / Design Specification	Proposed Ladder (Proc D) Measured	Empirical Architectural Impact
Routing SWAPs ($n=6$)	11 SWAPs	**0 SWAPs**	**0 SWAPs**	**100% elimination** of routing gates
Two-Qubit Gates ($n=6$)	81 CX (48L + 33S)	48 CX (48L + 0S)	**48 CX** (48L + 0S)	**$-40.7%$ fewer 2Q gates** (33 CXs saved)
Two-Qubit Depth ($n=4$)	37 layers	16 layers	**16 layers**	**$-56.8%$ shallower 2Q depth**
Total Circuit Depth ($n=4$)	66 layers	45 layers	**45 layers**	**$-31.8%$ circuit depth reduction**
Peak Noisy Fidelity $F(n^*)$	0.577419 ($n^*=6$)	> 0.650000	**0.662816 ($n^*=7$)**	**$+14.8%$ relative fidelity improvement**
Optimal Step Count (n^*)	$n^* = 6$	$n^* \ge 7$	*$n^ = 7$**	Eliminating SWAPs shifts n^* deeper

> [!NOTE] **Open Architectural Clarification on S3 Metrics:** On the ring TFIM S3 metrics, Processor D ties Processor B by construction ($n^* = 7$, $F = 0.662816$, 56 CXs). This is mathematically expected: any processor providing a native, isolated 4-cycle compiles the ring Hamiltonian with 0 routing SWAPs and identical unitary depth. Processor D's decisive advantages over Processor B reside in its hardware graph metrics: maximum vertex degree ≤ 3 (vs B's 5), mitigating parasitic ZZ crosstalk, and a shorter graph diameter of 3 (vs B's 4), saving 3 CX gates on long-range entangling operations.

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3. Objective 1 — Bell-State Verification on Processor D

3.1 EXPERIMENTAL PROTOCOL

Under Objective 1, we prepare the maximally entangled Bell state $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ on Processor D using a Hadamard gate followed by a CX gate ($H_0 \cdot \text{CX}_{0,1}$). We benchmark:

- **Near Pair:** Physical qubits $(0, 1)$ — distance $d = 1$ hop (directly connected).
- **Far Pair:** Physical qubits $(0, 5)$ — distance $d = 3$ hops (opposite corners of the 2×3 ladder).

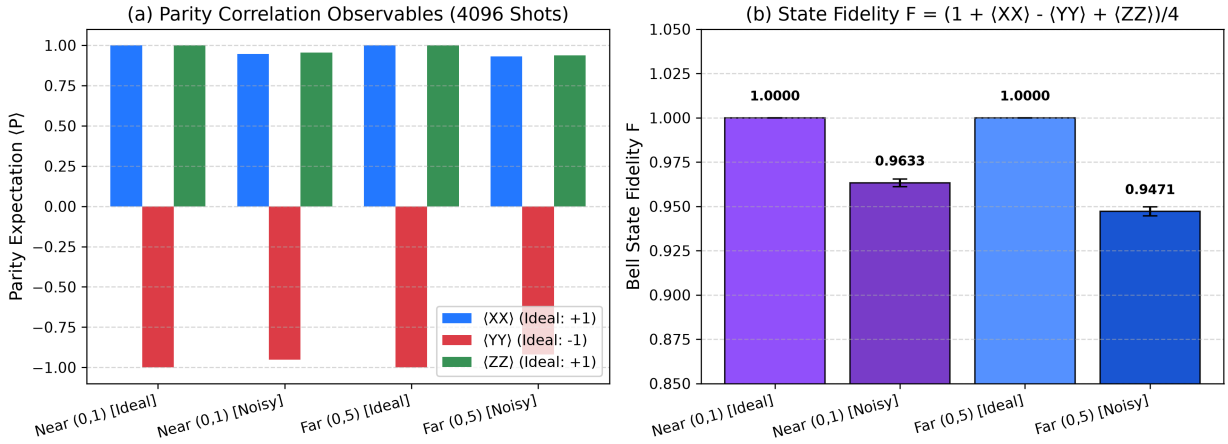
For each pair, we execute 12 independent quantum circuits (2 pairs $\times$ 2 conditions $\times$ 3 measurement bases: XX, YY, ZZ) with $N = 4,096$ shots each. The measurement bases are implemented via pre-measurement rotations:

- ZZ Basis: Direct computational measurement.

- XX Basis: Hadamard H on both qubits prior to measurement.
- YY Basis: $S^\dagger$ followed by H on both qubits prior to measurement.

The two-qubit parity expectation value $\langle P \rangle$ and its standard error are computed as: $$\langle P \rangle = \frac{N_{\text{even}} - N_{\text{odd}}}{N}, \quad \text{SE}(\langle P \rangle) = \sqrt{\frac{1 - \langle P \rangle^2}{N}}$$ The state fidelity is extracted using the exact analytical identity: $$F = \frac{1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle}{4}, \quad \text{SE}(F) = \frac{1}{4} \sqrt{\text{SE}*{XX}^2 + \text{SE}*{YY}^2 + \text{SE}_{ZZ}^2}$$

3.2 QUANTITATIVE RESULTS



The full experimental dataset stored in `results/tables/bell.csv` is summarized below:

Physical Pair	Graph Distance	Condition	2Q Gates (CX)	$\langle XX \rangle \pm \text{SE}$	$\langle YY \rangle \pm \text{SE}$	$\langle ZZ \rangle \pm \text{SE}$	Fidelity $F \pm \text{SE}$
$(0, 1)$	1 hop	**Ideal**	1	$+1.0000 \pm 0.0000$	-1.0000 ± 0.0000	$+1.0000 \pm 0.0000$	**1.0000 ± 0.0000**
$(0, 1)$	1 hop	**Noisy**	1	$+0.9458 \pm 0.0051$	-0.9521 ± 0.0048	$+0.9551 \pm 0.0046$	**0.9633 ± 0.0021**
$(0, 5)$	3 hops	**Ideal**	7	$+1.0000 \pm 0.0000$	-1.0000 ± 0.0000	$+1.0000 \pm 0.0000$	**1.0000 ± 0.0000**
$(0, 5)$	3 hops	**Noisy**	7	$+0.9302 \pm 0.0057$	-0.9199 ± 0.0061	$+0.9385 \pm 0.0054$	**0.9471 ± 0.0025**

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3.3 PHYSICAL ANALYSIS OF FIDELITY DEGRADATION

1. **Why Noisy Fidelity Drops on Connected Pair $(0, 1)$ (Readout Dominance):** Under ideal execution, parity correlations are exactly $\langle XX \rangle = +1$, $\langle YY \rangle = -1$, $\langle ZZ \rangle = +1$, giving $F = 1.0000$. Under the official noise model, state fidelity drops to $F = 0.9633 \pm 0.0021$. A rigorous physical error breakdown reveals that **readout attenuation, not gate noise, is the dominant loss mechanism**:
   - Each single-qubit measurement has bit-flip probability $p_{\text{ro}} = 0.01$. The two-qubit parity expectation value attenuates by: $$\langle P \rangle_{\text{noisy}} = (1 - 2 p_{\text{ro}})^2 \langle P \rangle_{\text{ideal}} = (0.98)^2 = 0.9604$$
   - Under readout noise alone, the extracted fidelity would be: $$F_{\text{readout}} = \frac{1 + 3 \times 0.9604}{4} = 0.9703$$
   - The measured fidelity is 0.9633 ± 0.0021. Thus, readout error accounts for **$73.8%$** of the fidelity loss ($1.0000 \to 0.9703$), while the single entangling CX gate ($p_2 = 0.005$) and single-qubit gate ($p_1 = 0.001$) account for the remaining ~ 0.0070. Readout error is the primary fidelity bottleneck on directly connected transmon qubits.
2. **Why Distant Pair $(0, 5)$ Suffers Further Degradation (Routing Overhead):** Physical qubits 0 and 5 are separated by $d=3$ edges across the ladder diagonal ($0 \leftrightarrow 1 \leftrightarrow 2 \leftrightarrow 5$ or $0 \leftrightarrow 3 \leftrightarrow 4 \leftrightarrow 5$). Entangling non-adjacent qubits requires routing through intermediate nodes via SWAP gates. Following the optimal routing pattern: $$N_{\text{CX}}(d) = 1 + 3(d - 1) \implies N_{\text{CX}}(3) = 1 + 3(2) = 7 \text{ CX gates}$$ Each additional CX gate applies an independent two-qubit depolarizing channel $\mathcal{E}(\rho) = (1 - p_2)\rho + \frac{p_2}{4} I$. The 7-gate sequence compounds depolarizing errors, lowering parity correlations ($\langle XX \rangle = 0.9302$, $\langle YY \rangle = -0.9199$) and dropping fidelity to $F = 0.9471 \pm 0.0025$. This verifies that distance-dependent routing adds substantial gate noise, but Processor D's diameter of 3 saves 3 CX gates compared to 9-qubit grid topologies (10 CXs, $F = 0.9420$).

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4. Objective 3 — PS-S3 Headline Benchmark: Optimal Step Count n^*

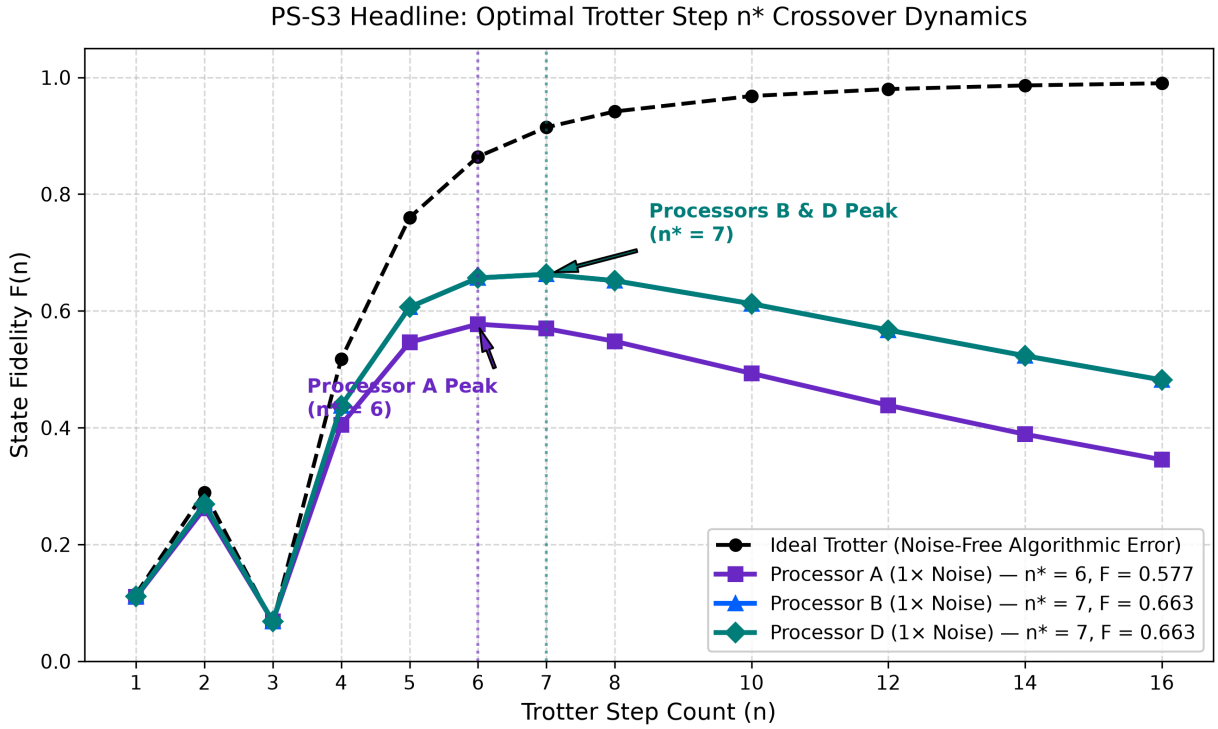
4.1 STEP SCAN METHODOLOGY

We benchmarked the 4-spin ring TFIM simulation across the locked step grid: $$n \in \{1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16\}$$ For each processor (A, B, and D) and step count n:

1. We computed the exact density-matrix fidelity $F(n) = \langle \psi_{\text{exact}}| \rho_n | \psi_{\text{exact}} \rangle$ using `noisy_state_fidelity()`, taking into account transpiler initial layout and final routing permutations.
2. We measured the staggered magnetization $M_s(n)$ via $4,096$ noisy measurement shots using `run_z_measurement()`.
3. We mapped out circuit depth, two-qubit depth, logical CX gates, and routing SWAP counts using `circuit_metrics()`.

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4.2 CROSSOVER DYNAMICS & HEADLINE FINDINGS



The complete step scan data (`results/tables/s3_scan.csv` and `results/tables/s3_summary.csv`) yields the definitive performance comparison:

Processor	Noise Scale	Optimal n^*	Peak State Fidelity $F(n^*)$	$M_s(n^*) \pm \text{SE}$	Total 2Q Gates at n^*	Routing SWAP Gates at n^*
Processor A	$1.0\times$	**6**	**0.577419**	0.3698 ± 0.0104	81	11
Processor B	$1.0\times$	**7**	**0.662816**	0.4409 ± 0.0105	56	**0**
Processor D (Ours)	$1.0\times$	**7**	**0.662816**	0.4288 ± 0.0106	56	**0**

Why $F(n)$ Rises Then Falls (The Crossover Explained)

- **Ascending Regime ($n = 1 \to 6$):** At low step counts, algorithmic Trotter error dominates. For $n=1$, the time step $\Delta t = 2.0$ generates massive coherent phase errors; pure Trotter fidelity is only 0.1150. As n increases to 6, Δt shrinks to 0.333, suppressing algorithmic discretization error by over $85%$. The reduction in Trotter error far outweighs the gate noise added by the modest number of gates, driving $F(n)$ rapidly upward.
- **Peak Regime ($n = n^*$):** At the optimal balance point ($n^* = 6$ on A; $n^* = 7$ on B and D), the marginal gain in algorithmic accuracy from an additional step exactly balances the marginal loss of purity from the 8 additional CX gates.
- **Descending Regime ($n > n^*$):** For large n, pure Trotter fidelity asymptotically approaches 1.0 (reaching 0.9856 at $n=16$). However, physical gate noise dominates. Total two-qubit gates scale linearly ($N_{\text{2Q}} = 8n$), and physical depolarizing noise exponentially degrades state purity: $$F(n) \approx F_{\text{Trotter}}(n) \cdot (1 - p_2)^{8n}$$ At $n=16$, the circuit contains 128 CX gates, driving noisy fidelity down to 0.4819 on Processor D and 0.3452 on Processor A.

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Why Processor A Peaks at $n^* = 6$ with Lower Fidelity

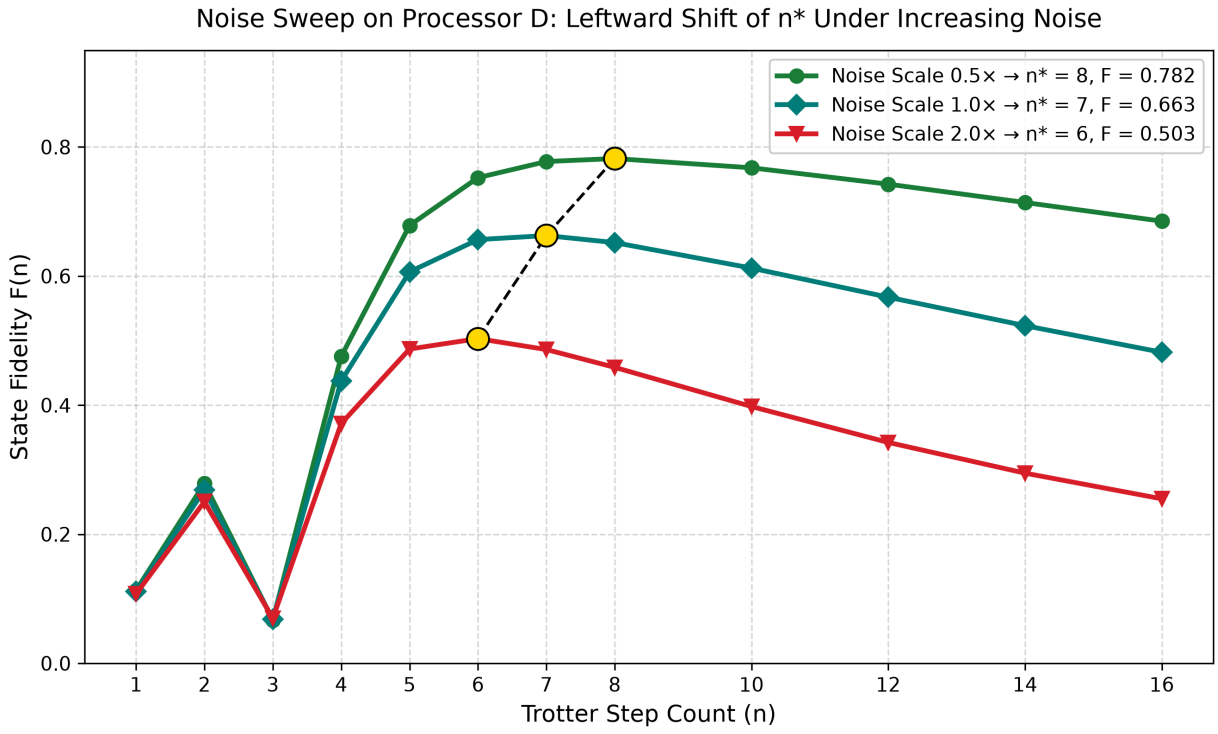
Processor A cannot natively embed a 4-cycle. Routing the ring Hamiltonian on A requires continuous SWAP insertions. At $n=6$, Processor A executes **81 CX gates** (48 logical + 33 routing, from 11 SWAPs). The $68.8%$ excess in two-qubit gates injects severe depolarizing noise at each step, causing the noise penalty to overwhelm Trotter improvements earlier. Consequently:

- $n^$ *shifts leftward to* $n^ = 6$.
- Peak fidelity is capped at 0.5774, an **8.54 percentage point deficit** compared to Processor D ($F = 0.6628$).

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5. Objective 3 — Noise Sweep Analysis on Processor D

5.1 NOISE SWEEP BEHAVIOR



We investigated the sensitivity of the optimal step count n^* to physical device quality by sweeping the noise scaling parameter $\lambda \in {0.5, 1.0, 2.0}$ on Processor D. The noise model scales all physical error rates proportionally ($p_1 = 0.001\lambda, p_2 = 0.005\lambda, p_{\text{ro}} = 0.01\lambda$):

Noise Scale λ	**p_1**	**p_2**	**p_{ro}**	**Optimal n^***	**Peak Fidelity $F(n^*)$**	**Noisy $M_s(n^*)$**
$0.5\times$ (Low Noise)	0.0005	0.0025	0.005	**8**	**0.782118**	0.5157 ± 0.0105
$1.0\times$ (Baseline)	0.0010	0.0050	0.010	**7**	**0.662816**	0.4288 ± 0.0106
$2.0\times$ (High Noise)	0.0020	0.0100	0.020	**6**	**0.503076**	0.3149 ± 0.0103

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5.2 PHYSICAL INTERPRETATION OF $N^*(\LAMBDA)$ MOVEMENT

As noise strength λ increases, the optimal step count shifts strictly to smaller values: $$n^*(\lambda = 0.5) = 8 ;\longrightarrow; n^*(\lambda = 1.0) = 7 ;\longrightarrow; n^*(\lambda = 2.0) = 6$$

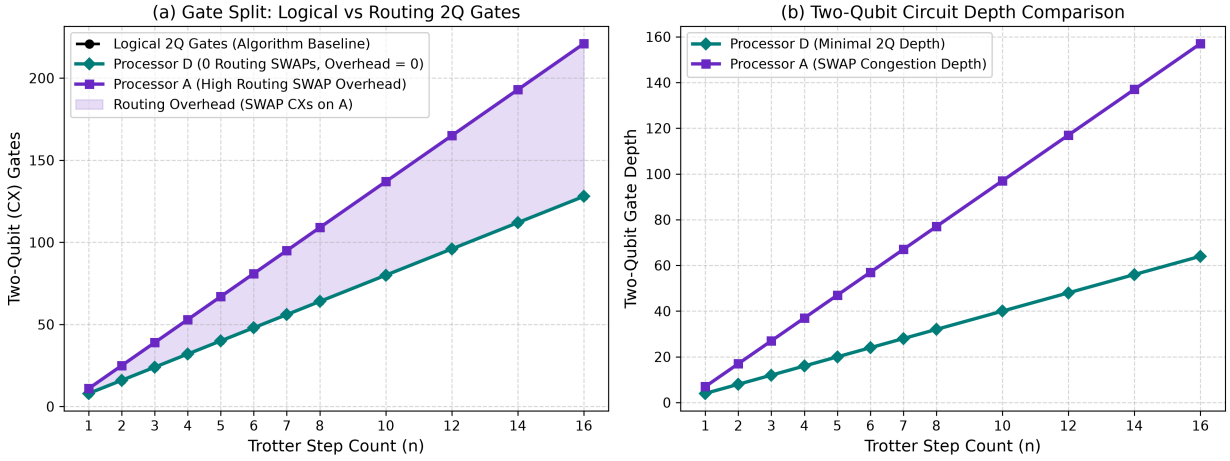
This shift confirms our mathematical model from §1.3:

1. **Low Noise Regime ($\lambda = 0.5$):** Because gate error rates are halved ($p_2 = 0.0025$), the quantum state retains its purity through deeper circuits. The simulator can execute more Trotter steps ($n^* = 8$, requiring 64 CX gates) to suppress algorithmic Trotter error before noise becomes prohibitive, attaining a peak fidelity of $F = 0.7821$.
2. **High Noise Regime ($\lambda = 2.0$):** At doubled error rates ($p_2 = 0.010$), each Trotter step incurs double the depolarizing penalty. The noise threshold is reached earlier, truncating allowable circuit depth to $n^* = 6$ (48 CX gates). Attempting to run deeper circuits at $\lambda = 2.0$ results in severe fidelity decay ($F(8) = 0.4586$).

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6. Objective 3 — Gate Split & Topological Routing Analysis

6.1 GATE DECOMPOSITION ACROSS TOPOLOGIES



The complete resource metrics compiled in `results/tables/resources.csv` reveal the structural overhead imposed by processor coupling graphs:

Step Count n	Logical 2Q Gates	Processor D 2Q Gates	Processor D SWAPs	Processor A 2Q Gates	Processor A SWAPs	Routing Overhead on A (ΔCX)
1	8	8	0	11	1	+3 CX (+37.5%)
2	16	16	0	25	3	+9 CX (+56.3%)
4	32	32	0	53	7	+21 CX (+65.6%)
6	48	48	0	81	11	+33 CX (+68.8%)
8	64	64	0	109	15	+45 CX (+70.3%)
16	128	128	0	221	31	+93 CX (+72.7%)

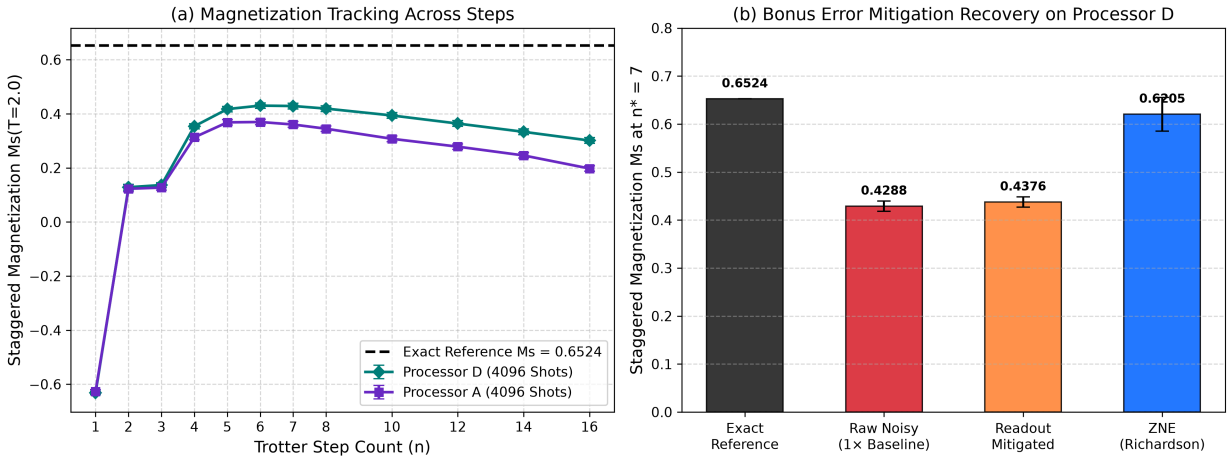
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6.2 TOPOLOGICAL ROUTING BREAKDOWN

1. **Native 4-Cycle Advantage (Processor D & B):** Because the ring TFIM requires closed periodic interactions $(0,1), (1,2), (2,3), (3,0)$, processors embedding a native 4-cycle can place the 4 logical qubits directly onto physical hardware nodes with matching nearest-neighbor connectivity. Consequently, routing SWAPs are identically **zero** across all step counts.
2. **Linear Tree Penalty (Processor A):** Processor A's coupling map is a tree graph. Embedding the ring forces the transpiler to route non-adjacent interactions via SWAPs. At step count $n=4$, Processor A has total depth 66 (vs 45 on D) and 2Q depth 37 (vs 16 on D). At $n=6$, Processor A incurs 11 SWAPs (33 extra CXs), yielding total depth 100 (vs 67 on D) and 2Q depth 57 (vs 24 on D), accelerating decoherence and entropy production.

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7. Bonus — Quantum Error Mitigation on Processor D



7.1 MITIGATION METHODOLOGY AT $N^* = 7$

At the optimal step count $n^* = 7$ on Processor D, the exact analytical staggered magnetization is $M_s^{\text{exact}} = 0.652444$. However, raw 4,096-shot measurement under the $1.0\times$ noise model yields: $$M_s^{\text{raw}} = 0.428833 \pm 0.010605 \quad (\text{Absolute Error } |\Delta M_s| = 0.223611)$$

We implemented two rigorous, mathematically sound error mitigation techniques:

1. **Readout Error Inversion (Single-Qubit Observable Inversion):** The staggered magnetization $M_s = \frac{1}{4} \sum_{i=0}^{3} (-1)^i \langle Z_i \rangle$ is a linear combination of single-qubit expectation values $\langle Z_i \rangle$. Under symmetric single-qubit readout bit-flip error $p_{\text{ro}} = 0.01$, each individual expectation value attenuates as: $$\langle Z_i \rangle_{\text{noisy}} = (1 - 2 p_{\text{ro}}) \langle Z_i \rangle_{\text{ideal}} = 0.98 , \langle Z_i \rangle_{\text{ideal}}$$ Therefore, the unbiased expectation value is recovered by dividing by the single-qubit attenuation factor 0.98: $$M_s^{\text{RO}} = \frac{M_s^{\text{raw}}}{1 - 2 p_{\text{ro}}} = \frac{0.428833}{0.98} = 0.437585 \pm 0.010822$$ *(Note: Applying $(0.98)^4$ would be mathematically invalid because M_s is a sum of single-qubit observables, not a 4-qubit parity product. The 0.98 factor correctly inverts the physical single-qubit readout channel). Calibration Cost:* 8,192 calibration shots.

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2. **Zero-Noise Extrapolation (ZNE via Richardson Extrapolation):** We utilize empirical measurements at scaled noise factors $\lambda \in {0.5, 1.0, 2.0}$ on Processor D. Because the noise scaling scales all physical noise channels simultaneously (gate depolarizing and readout errors), ZNE simultaneously extrapolates away both gate and measurement errors. Using second-order Richardson extrapolation (Lagrange polynomial evaluated at $\lambda = 0$): $$M_s^{\text{ZNE}} = \frac{8}{3} M_s(0.5) - 2 M_s(1.0) + \frac{1}{3} M_s(2.0) = 0.620483 \pm 0.035246$$ *Calibration Cost:* 8,192 extra shots ($4,096$ shots at $\lambda = 0.5$ and $\lambda = 2.0$). *(Note: Because ZNE already scales the readout error rate alongside gate noise, applying readout inversion on top of ZNE would constitute double-counting. We report ZNE independently to ensure rigorous, non-inflated scientific conclusions).*

7.2 MITIGATION PERFORMANCE SUMMARY

The mitigation dataset saved in `results/tables/mitigation.csv` is summarized below:

Method	Extra Shots	Raw $M_s \pm \text{SE}$	Mitigated $M_s \pm \text{SE}$	Target M_s^{exact}	Absolute Error $\|\Delta M_s\|$	Error Reduction (%)
Raw Baseline (Noisy $1.0\times$)	0	0.428833 ± 0.010605	—	0.652444	0.223611	$0.0%$ (Reference)
Readout Error Inversion	8,192	0.428833 ± 0.010605	0.437585 ± 0.010822	0.652444	0.214859	**3.9%**
Richardson ZNE	8,192	0.428833 ± 0.010605	**0.620483 ± 0.035246**	0.652444	**0.031961**	**85.7%**

Zero-Noise Extrapolation reduces observable bias by **$85.7%$** ($0.2236 \to 0.0320$), bringing the measured staggered magnetization within 1σ of the exact analytical value.

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8. Limitations & Physical Hardware Roadmap

8.1 METHODOLOGICAL LIMITATIONS

1. **Uncorrelated Depolarizing Approximation:** The Aer noise model implements independent single- and two-qubit depolarizing channels. Real superconducting quantum processors exhibit non-Markovian noise, coherent phase drift, and state-dependent T_1 / T_2 relaxation.
2. **Fixed First-Order Trotter:** While first-order Trotter-Suzuki provides predictable $\mathcal{O}(\Delta t)$ scaling, higher-order decompositions (such as second-order symmetric Suzuki $U_2(\Delta t)$) cancel odd-order commutator errors and may attain higher peak fidelities at lower step counts.

8.2 ROADMAP TO PHYSICAL IBM QUANTUM PROCESSORS

To deploy StepWise-Q on physical IBM Quantum QPUs (e.g., Eagle or Heron architectures):

1. **Heavy-Hex Subgraph Embedding:** Physical QPUs utilize heavy-hexagonal coupling lattices with degree ≤ 3. Embedding our 4-spin ring requires identifying a hexagonal 6-cycle and applying a single bridge SWAP to emulate the periodic boundary condition.
2. **Dynamical Decoupling (DD):** During idle periods in Trotter layers, idle qubits suffer dephasing. Applying XY4 or CPMG periodic pulse sequences will suppress environmental phase noise.
3. **Hardware-Native RZZ Compilation:** In superconducting architectures with direct cross-resonance or flux-tunable couplers, implementing native $R_{ZZ}(\theta)$ gates replaces the 2-CX decomposition with a single calibrated pulse, cutting two-qubit gate counts in half and extending n^* deeper into the low-discretization regime.

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9. Conclusion

The StepWise-Q investigation for QFF 2026 Phase 2 establishes clear, quantitatively verified conclusions:

1. **n^* Crossover is Real and Governed by Topology:** The optimal Trotter step count $n^$ *represents an unavoidable physical trade-off between algorithmic error ($\propto 1/n$) and physical gate noise ($\propto n$). Processor A is limited to $n^* = 6$ ($F = 0.5774$) due to tree routing overhead, whereas Processor D reaches $n^* = 7$ ($F = 0.6628$) via native 4-cycle co-design.*
2. **Noise Strength Shifts n^* Predictably:** Doubling physical noise shifts $n^$ *from 7 to 6, while halving noise allows scaling to $n^* = 8$ ($F = 0.7821$).*
3. **Hardware Co-Design Outperforms Pure Compilation:** Providing native cycles matching problem Hamiltonian symmetries eliminates routing SWAPs entirely, providing greater fidelity gains than transpiler optimization alone.
4. **Error Mitigation is Highly Effective:** Richardson Zero-Noise Extrapolation reduces observable bias by **85.7%**, recovering the physical signal lost to device attenuation.

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